Kelan S. Zielinski

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Work & Leadership Experience

Motorola Solutions Inc. **Plantation, Fl**
Electrical Engineering Intern *May 2024 - August 2024*

- Successfully achieved optimal impedance matching for 109.65MHz crystal filters using a Smith chart to analyze and adjust circuit parameters.
- Ensured signal integrity by implementing automated tests in LabVIEW to verify network matching, comparing RSSI values to predetermined benchmarks.
- Validated scattering (S2P) parameters on a Vector Network Analyzer (VNA) and redesigned the network on Advanced Design System (ADS) software.
- Identified timing issues causing unexpected power cycles between a radio and UPS using an oscilloscope, enabling a targeted solution that improved system reliability and reduced troubleshooting time.

IoT-Driven Crop Production System **Coral Gables Fl**
Hardware Engineering Intern - University of Miami *January 2023 - May 2024*

- Designed and built an IoT-controlled hydroponics system to cultivate 2000+ plants per year within a footprint of less than 50 square feet.
- Developed a hardware prototype and validated components, including I2C and UART sensors and actuators, based on system requirements.
- Utilized an EDA simulation platform to model and optimize system circuitry for improved performance and reliability.

University Of Miami – Design For Testability Laboratory **Coral Gables, Fl**
Teaching Assistant *January 2024 - May 2024*

- Assisted students in designing and implementing a digital signal processor (DSP) using VHDL.
- Supported students in troubleshooting and resolving timing issues in complex digital systems, ensuring proper functionality and performance.

HKN Honor Society (IEEE) – University of Miami Chapter **Coral Gables, Fl**
Eta Kappa Nu – Honor Society Member *February 2024 - Present*

- Supported 30+ students in mastering fundamental engineering concepts.

Projects

FPGA-Based Quantum Emulator – VHDL, Python, ModelSim **May 2025**

- Designed a physics-based quantum system on an FPGA, modeling the state evolution of qubits.
- Leveraged FPGA mass-parallelism to execute matrix exponentiation in constant time complexity(O(1)).
- Developed a user-friendly GUI to seamlessly communicate with the FPGA, enabling real-time control of emulation parameters and visualization of emulation results.

32-bit MIPS RISC CPU – VHDL, ModelSim, Quartus Prime **October 2024**

- Designed and implemented a modular 32-bit MIPS-based RISC processor in VHDL.

Advanced Digital Signal Processor – VHDL, ModelSim **April 2024**

- Implemented the ADSP-2100 digital signal processor to perform specific computations based on the instruction.
- Integrated Built-In-Self-Test (BIST) architecture to generate signatures and expose faults within the system design.

Rotor-based Encryption Machine – Verilog, ModelSim, Quartus Prime **November 2023**

- Developed a Verilog-based model of the Enigma M3, featuring three dynamic rotors with rotational functionality and a reflector component.
- Verified data transfer and timing aspects of system design using RTL viewer and waveforms.

Education & Technical Skills

University of Miami **Coral Gables, Fl**
Bachelor of Science in Electrical Engineering – GPA: 3.72 *May 2025*

- **Relevant Coursework:** Computer Organization & Design, Digital Signal Processing(DSP), Discrete Time Signals & Systems, Communication Systems, Circuits Signals & Systems, Design for Testability(DFT), Electronics, Logic Design, Structured Digital Design, Data Structures

Technical Skills: VHDL, Verilog, C++, C, Python, ARM Assembly, Soldering, PCB Design & Testing
Developer Tools: ModelSim, ADS, Cadence, PSpice, MATLAB, Intel Quartus Prime, KiCad, Unix, LabVIEW, AutoCAD
Test Equipment: Oscilloscopes, Spectrum Analyzers, Logic Analyzers, Network Analyzers, Multimeters, Signal Generators